

μ Touch: Enabling Accurate, Lightweight Self-Touch Sensing with Passive Magnets

Siyan Wang*, Ke Li[†], Jingyuan Huang*, Jike Wang*, Cheng Zhang[†], Alanson Sample[‡], Dongyao Chen*

*Shanghai Jiao Tong University, China

[†]Cornell University, USA

[‡]University of Michigan, USA

Email: {wsy0227, aquamarine_indigo, jikewang, chendy}@sjtu.edu.cn
{kl975, chengzhang}@cornell.edu; apsample@umich.edu

QUICK LINKS

Code repo	github.com/Wangmerlyn/muTouch
MagX base	github.com/dychen24/magx
Models	Release tag backup/3_dim-models-20260121 on GitHub Releases (Assets)
PCB sources	pcb/ (muTouch Altium project; legacy Magway.* filenames)

I. SCOPE

This guide describes the artifact supporting μ Touch: Enabling Accurate, Lightweight Self-Touch Sensing with Passive Magnets [1]: hardware (muTouch PCB + magnets) and software (BLE data collection, semi-supervised classifier). It targets reviewers who want to install, run, and validate the pipeline. Component specs and how they are verified: hardware specs and outputs in `pcb/README.md` (used in Section V); firmware/data-collection specs and expected signals in `Codes/read_raw_ble/README.md` (used in Sections V–VI); project overview and folder purposes (including artifact/) in `README.md`.

II. NOTATION

Paths are relative to the repo root unless stated; runtime scripts assume `Codes/` as the working directory.

III. BILL OF MATERIALS & REQUIREMENTS

A. Hardware (minimal)

- muTouch PCB (Altium project in `pcb/`; assembled board; filenames use legacy `Magway.*`). PCB design by Xiaomeng Chen.
- 1–2 passive N52 grade magnets (6–8 mm recommended).
- Host laptop: Ubuntu 20.04+ or macOS 12+, 4-core CPU, ≥ 8 GB RAM, BLE 4.0+ adapter.
- Optional: BLE USB dongle (if desktop lacks BLE).

B. Software

- Python 3.10; Conda virtual env named `muTouch` recommended.
- Git with submodules; CMake/Make only if you rebuild the C++ solver.

- Dependencies installed via `pip install -r Codes/requirements.txt`.
- LaTeX/PDF tools only needed to rebuild this document.

IV. OBTAINING THE ARTIFACT

1) Clone the repository (now public):

```
REPO_SSH=git@github.com:Wangmerlyn/muTouch.git
git clone --recurse-submodules "$REPO_SSH"
# HTTPS fallback:
# REPO_HTTPS=https://github.com/Wangmerlyn/muT
# git clone --recurse-submodules "$REPO_HTTPS"
```

If you hit SSH “permission denied”, use the HTTPS command above or ensure your SSH key is added to GitHub.

2) Activate env:

```
conda create -n muTouch python=3.10
conda activate muTouch
```

You should see the shell prompt prefixed with (`muTouch`) after activation; this isolates dependencies for repeatability.

3) Install deps:

```
pip install -r Codes/requirements.txt
```

Expect a clean install without errors; rerun from the repo root if paths are not found.

4) Models: snapshot tag

```
backup/3_dim-models-20260121.
```

Download binaries from GitHub Releases (Assets).

5) Set working directory for runtime scripts to `Codes/`:

```
cd Codes
```

V. SETUP & CONFIGURATION

1) Flash firmware:

```
Codes/Arduino/bleReadMultiple/bleReadMultiple.ino in
```

Arduino IDE;

select Feather nRF52; upload.

- 2) Find BLE address: `python read_raw_ble/find_device.py` (copy device MAC/UUID).
- 3) Hardcode BLE address: edit the `address = ''...''` line near the bottom of:
 - `read_raw_ble/read_sensor.py`
 - `read_raw_ble/read_sensor_real.py`
 - `read_raw_ble/read_sensor_real_classifier.py` (if used)
- 4) Calibration capture (run inside Codes/):

```
python read_raw_ble/read_sensor.py
```

Do a brief figure-8 motion away from metal surfaces; CSVs are saved under `datasets/`.

- 5) Offsets/scales: place the latest files named `offset-*.npy` and `scale-*.npy` in `calibration_files/`. The scripts automatically load the newest files with those prefixes.
- 6) Models: ensure `read_raw_ble/models/` holds the downloaded checkpoint set if you need pretrained classifiers.

VI. RUNNING THE ARTIFACT

A. Data capture

Captures raw magnetometer streams over BLE with the latest calibration, writing timestamped CSVs under `datasets/`.

```
SCRIPT=read_raw_ble/read_sensor_real.py  
python "$SCRIPT"
```

B. Real-time classification

Runs the live classifier; expect gesture labels printed to console and logged in `datasets/`.

```
SCRIPT=read_raw_ble/read_sensor_real_classifier.py  
python "$SCRIPT"
```

Ensure the script uses the latest `offset-*`, `scale-*`, and model files (picked automatically if they sit in `calibration_files/` and `read_raw_ble/models/`).

C. Expected outcomes

- Face-touching: $\approx 93\%$ accuracy (8 gestures) with 3 s fine-tuning/user.
- Scratch detection: $\approx 95\%$ accuracy (9 gestures) with 3 s fine-tuning/user.
- Real-time loop maintains $>30\text{Hz}$ inference on a laptop CPU.

VII. REPRODUCIBILITY CHECKLIST

- **Hardware reproducible:** PCB sources + BOM (muTouch; files named `Magway.*` for compatibility) included.
- **Software reproducible:** All scripts + TS2Vec submodule; pinned deps in `Codes/requirements.txt`.
- **Data:** Calibration and small demo runs can be generated locally; full datasets are participant-specific and not included.
- **Pretrained models:** Provided via GitHub tag `backup/3_dim-models-20260121`.

VIII. TROUBLESHOOTING

- BLE not found: retry `find_device.py`; check power and pairing blocks; use BLE dongle.
- Drifting predictions: recalibrate sensors; ensure distance from large metal; re-run offset/scale.
- Import errors: confirm submodule init (`git submodule update --init --recursive`) and Python path from repo root.

IX. FOR COMPLETE INSTRUCTIONS

Due to page limits, this PDF is a concise checklist. For full, continuously updated steps and tips, please refer to the repository `README.md` and the Markdown guide (best viewed via GitHub preview): `artifact/README.md`. Use those as the authoritative references for reproduction and further code tweaks.

X. NOTES ON PRIOR WORK

The project builds on MagX (MobiCom'21) codebase for magnetic sensing [2]. This artifact extends it to self-touch sensing and includes updated PCB by Xiaomeng Chen.

REFERENCES

- [1] S. Wang, K. Li, J. Huang, J. Wang, C. Zhang, A. Sample, and D. Chen, " μ touch: Enabling accurate, lightweight self-touch sensing with passive magnets," in *Proceedings of the IEEE International Conference on Pervasive Computing and Communications (PerCom)*, Pisa, Italy, Mar. 2026, to appear.
- [2] D. Chen, M. Wang, C. He, Q. Luo, Y. Iravantchi, A. Sample, K. G. Shin, and X. Wang, "Magx: Wearable, untethered hands tracking with passive magnets," in *Proceedings of the 27th Annual International Conference on Mobile Computing and Networking (MobiCom '21)*, 2021, pp. 269–282.